

A Multi-Vehicle Self-Organized Cooperative Control Strategy for Platoon Formation in Connected Environment

Mengqi Zhang¹, Chunyan Wang¹, Wanzhong Zhao¹, Jinqiang Liu¹, and Ziyu Zhang¹

Abstract—Aiming at the problem of low merging efficiency, poor platoon stability, and high collision risk when multiple connected and automated vehicles merge into the same target platoon, we propose a multi-vehicle self-organized cooperative control strategy for platoon formation, which includes vehicle self-organizing formation control and platoon cooperative merging control. The vehicle self-organizing formation control module organizes the merging vehicles within the V2V communication range into multiple local platoons. The dynamic self-adjusting critical interval and a fixed-topology second-order platoon consistency control protocol are proposed to divided the vehicles reasonably and make the states of local platoon vehicles consistent. The merging vehicles merge into the target platoon as a whole in the form of a local platoon, which transforms the complex multi-vehicle merge problem into a platoon cooperative control problem and improves the merging efficiency. The platoon cooperative merging control module adopts a distributed model predictive control (DMPC) theory to design two longitudinal cooperative merging controllers, which control the target platoon to split to create a merging gap and the local platoon to align with this gap longitudinally. The lateral merging controller controls the local platoon change the lane to merge into the target platoon safely and smoothly. Simulation experiments are conducted in typical scenarios, and it is verified that the proposed control strategy can enable multi-vehicles to merge into a platoon efficiently, safely, and stably.

Index Terms—Connected and automated vehicles, multi-vehicle cooperative control, platoon formation and control, local platoon, distributed model predictive control.

I. INTRODUCTION

THE development of Connected and Automated Vehicle (CAV) technology provides a new mode for vehicle driving, namely the platoon-based driving mode. Platooning has attracted much attention due to its advantages of improving traffic efficiency, reducing accident rates, and alleviating traffic

congestion [1], [2], [3]. The structure of the platoon is not fixed in the driving process. Flexible platoon expansion needs to respond to the dynamic merging requirements of vehicles. Therefore, in the design of the platoon system, it is an important goal to control vehicles to merge into the platoon efficiently, safely, and stably [4]. The vehicle merging process requires dynamic adjustment between the platoon vehicles and the merging vehicles, and at the same time, the safety and stability of the platoon should be ensured, which puts forward strict requirements for the longitudinal control of vehicles merging into a platoon [5]. When multiple vehicles merge into the same platoon, the problem becomes more complicated.

In traffic, irregular roadways and roadway transition areas are the major sections of the roadways that generate the vehicle merging process. In the vicinity of typical lane reduction roadways, such as the construction of a portion of the lane ahead, a traffic accident in a lane ahead, or a lane reduction due to topographic factors, vehicles originally traveling in the terminated lane will need to change lanes to merge into the continuous lane before reaching the end of the lane. In this case, multiple vehicles traveling in the terminated lane will simultaneously make requests to merge into the vehicle platoon traveling in the continuous lane. Without a proper control strategy to guide the behavior of the merging vehicles and the platoon, it is easy to create traffic accidents and jams at the lane bottleneck. Therefore, how to control multiple merging vehicles to merge into the target platoon efficiently, safely and stably is a valuable research topic.

Some existing researches has proposed a variety of solutions to the problem of multi-vehicle merging into the platoon and the development of platoon cooperative control has also provided new ideas for this problem. However, there are still some challenges in terms of vehicle merging methods and platoon merging control. On the one hand, the existing researches decompose the multi-vehicle merging problem into multiple single-vehicle merging problems, which will lead to multiple splits in the target platoon and low merging efficiency [4]. However, organizing the merging vehicles into local platoons and merging them into the target platoon as a whole exists the challenges of quickly making the states of local platoon vehicles consistent. Therefore, how to reasonably divide the merging vehicles to form local platoons and quickly adjust the states of the platoon members within a limited distance to achieve consistency is a major difficulty that this paper needs to solve. On the other hand, the existing methods mostly

Received 10 June 2024; revised 11 September 2024 and 4 November 2024; accepted 17 November 2024. Date of publication 29 January 2025; date of current version 3 March 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 52372390, in part by the Fundamental Research Funds for the Central Universities under Grant NC2022002 and Grant NI2023002, in part by Jiangsu Key Research and Development Plan under Grant BE2022053, and in part by the Postgraduate Research and Practice Innovation Program of Jiangsu Province under Grant KYCX22_0338. The Associate Editor for this article was R. Malekian. (Corresponding author: Chunyan Wang.)

The authors are with the Department of Vehicle Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing 210016, China, and also with Jiangsu Engineering Research Center of Vehicle Distributed Drive and Intelligent Wire Control Technology, Nanjing 210016, China (e-mail: zmq0219@nuaa.edu.cn; wangcy@nuaa.edu.cn; zzw@nuaa.edu.cn; jinqiang916@163.com; zhangziyu@nuaa.edu.cn).

Digital Object Identifier 10.1109/TITS.2024.3504572

1558-0016 © 2025 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See <https://www.ieee.org/publications/rights/index.html> for more information.

focus on the precise control of vehicle positions during the merging process while ignoring the velocity changes of the platoon vehicles. When the initial velocity of platoon vehicles is different, it is easy to cause serious velocity fluctuations, and the stability of the platoon deteriorates. Therefore, how to control the velocity of vehicles to change smoothly and ensure the stability of the platoons during the merging process is another difficulty that this paper needs to solve.

To overcome the above difficulties, we propose a multi-vehicle self-organized cooperative control strategy for platoon formation. Specifically, a vehicle self-organizing formation control strategy for merging vehicles is proposed, which reasonably divides the merging vehicles to form local platoons and quickly makes the states of local platoon vehicles consistent within a limited distance. Then, two DMPC-based distributed longitudinal cooperative merging controllers and a lateral merging controller are designed. The velocity fluctuations are suppressed by designing the optimization problems to ensure the platoon stability in the merging process. The main contributions of this paper are as follows:

- 1) A multi-vehicle self-organized cooperative control strategy is proposed. The merging vehicles self-organize into local platoons and merge into the target platoon as a whole. Compared to existing algorithms, the merging efficiency and platoon stability are significantly improved.

- 2) Propose a vehicle self-organizing formation control strategy based on dynamic self-adjusting critical interval and second-order platoon consistency control protocol to transform the complex multi-vehicle merge problem into a platoon cooperative control problem and improve the merging efficiency.

- 3) Two DMPC-based longitudinal cooperative merging controllers are designed, which consider the control problems of local platoon and target platoon as two types of constrained optimization problems with different goal orientations. It avoids serious velocity fluctuations of the vehicles and effectively improves the safety and stability of the merging process.

The remainder of this article is organized as follows. Section II is a brief summary of the research related to this article. Section III proposes a framework for the multi-vehicle self-organized cooperative control strategy. Section IV proposes a vehicle self-organizing formation control. Section V designs the distributed cooperative merging controllers. The simulation results and analysis are shown in Section VI. In Section VII, this study is concluded.

II. LITERATURE REVIEW

This section summarizes the previous literature. Section II-A introduces the platoon formation strategies. Section II-B illustrates the vehicle merging control strategies including the single vehicle merging strategies and multi-vehicle merging strategies. Section II-C describes the platoon cooperative control strategies.

A. Platoon Formation Strategies

Platoon control algorithms and communication systems are necessary conditions for platoon formation. Adjusting the state

of the platoon members to achieve consistency and ensuring the stability of the platoon traveling process are the main functions that the platoon control algorithm needs to realize. Many platoon formation algorithms have been proposed so far. CACC control strategy is one of them. Ploeg et al. proposed a particular CACC control strategy that receives much attention and this controller design is experimentally validated [6]. Ploeg et al. also studied string stability of vehicle platoons based on distributed consistency control algorithms and gave a stricter definition of string stability [7]. Guo et al. proposed a new distributed adaptive integrated-sliding-mode controller to ensure the finite-time stability and string stability of vehicle platoons [8]. Lan et al. proposed a distributed min-max robust model predictive control algorithm with V2V communication delay to ensure the asymptotic stability of vehicle platoons [9]. Although many novel platoon control algorithms have been proposed, CACC is still commonly used because of its flexibility and effectiveness [12].

B. Vehicle Merging Control Strategies

The simplest scenario of vehicles merging into a platoon is a single-vehicle merging, for which various control strategies have been proposed by different research works. An et al. proposed a platoon gap control method based on hybrid ACC-DMPC, which laid the foundation for vehicle merging [10]. Cheng et al. and Scholte et al. realized the process of a free-vehicle merging into the platoon based on the virtual target control strategy (VTCS) [11], [12]. Maiti et al. investigated the impact of different merging positions on traffic efficiency [13].

Based on the research on single vehicle merging into a platoon, some scholars decomposed the multi-vehicle merging problem into multiple single-vehicle merging problems. Kazerooni et al. divided merging vehicles and platoon vehicles into multiple triples, and the single-vehicle interaction protocol is implemented in each triple [14]. In the method proposed by Hidalgo et al., the platoon members open multiple gaps for the merging vehicles at the same time [15]. Ding et al. and Liu et al. proposed rule-based and sequence-based cooperative merging strategies, which achieves a good balance between computation and traffic efficiency [16], [17]. Wu et al. extended the research scenario from single-lane or dual-lane to multi-lane and organized vehicles in multiple lanes into a platoon [18]. The above methods of applying the strategies of single-vehicle merging into a platoon to the multi-vehicle merging scenario cause multiple splits in the target platoon and low merging efficiency.

C. Platoon Cooperative Control Strategies

At present, the cooperative control of platoons has been paid attention to by more and more researchers, among which the platoon merging control is an important research field. Liu et al. proposed provably-correct self-organizing strategies. The merging of multiple different platoons is achieved through a provably-correct safety layer [19]. Mu et al. developed an event-triggered rolling horizon-based systematical trajectory

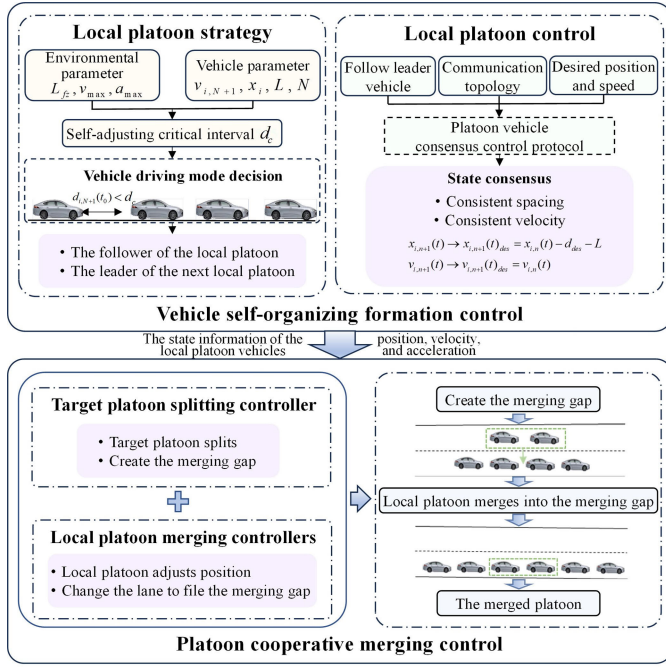


Fig. 1. The framework of multi-vehicle self-organized cooperative control strategy for platoon formation.

planning (RSTP), which enabled the safe and smooth merging of two platoons within the trajectory control area [20]. Xue et al. projected a platoon from one lane to another to form a virtual platoon, thereby achieving platoon merging [21]. Morales Medina et al. also adopt the strategy of virtual vehicle platoons. Different from Xue, they mainly aimed at intersection scenarios [22]. Xiong et al. and Deng et al. also studied the cooperation of vehicle platoons at highway intersections to minimize the time cost and fuel consumption when platoon merging [23], [24]. Karbalaieali et al. proposed an automatic merging control algorithm to guide ramp vehicles to merge effectively [25]. Charitidou et al. proposed a platoon merging method based on signal temporal logic and the solution of convex quadratic programming is used as the control input of the vehicles [26]. Min et al. designed a DMPC2 controller, which can control two platoons to merge in different scenarios [27].

Hence, converting the multi-vehicle merging problem into a platoon cooperative control problem helps to reduce the complexity of the problem. Motivated by the above researches, this paper proposes a multi-vehicle self-organized cooperative control strategy for platoon formation, which includes vehicle self-organizing formation control and platoon cooperative merging control.

III. FRAMEWORK

As shown in Fig. 1, we propose a multi-vehicle self-organized cooperative control strategy for platoon formation to solve the problem of multi-vehicle merging into the same target platoon. Fig. 2 shows a scenario where multi-vehicle merge into the target platoon cooperatively. The target platoon is driving in the right lane, and the merging vehicles are driving in the left lane. Under connected environments, vehicles exchange information through vehicle-to-vehicle (V2V)

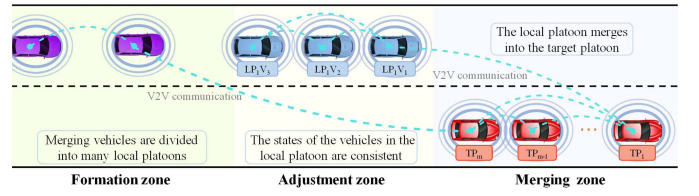


Fig. 2. Scene diagram of multi-vehicle merging into the target platoon.

communication technology, and each vehicle can obtain the position, velocity, and acceleration of the vehicle it communicates with. The whole merging scenario consists of three parts: the formation zone, the adjustment zone and the merging zone, and their lengths are determined by the coverage of the V2V communication technology [28].

In Fig. 1, the vehicle self-organizing formation control module organizes the merging vehicles into multiple local platoons and the entire local platoon merges into the target platoon to improve the merging efficiency. After entering the formation zone, the merging vehicles send out an application to merge into the platoon, and then they are organized into multiple local platoons LP_i according to the dynamic self-adjusting critical interval. The vehicles in the local platoon i are denoted by $LP_i V_n (n = 1, \dots, N)$, $n = 1$ represents the leading vehicle of the local platoon [29]. After the local platoon enters the adjustment zone, the platoon vehicle consistency control protocol controls the leading vehicle and the following vehicles to adjust their states respectively, so that the longitudinal spacing and velocity of the vehicle in the local platoon converge to be consistent.

After the local platoon enters the merging zone, it merges into the target platoon TP as a whole. The vehicles in the target platoon are denoted by $TP_m (m = 1, \dots, M)$, $m = 1$ represents the leading vehicle of the target platoon. The platoon cooperative merging control module in Fig. 1 designs two longitudinal cooperative merging controllers by distributed model predictive control algorithm and a lateral merging controller. By solving two types of bounded constrained optimization problems with different goal orientations, smooth and feasible longitudinal optimal control is obtained. The first optimization problem considers the safe space required for the merging process and controls the target platoon splitting to create the platoon gap. Another optimization problem controls the local platoon to accurately align with the platoon gap. At the same time, the two optimization problems are designed with the difference in the velocity of the platoons in mind, and velocity adjustment factors and input constraints are introduced to avoid large velocity fluctuations of the vehicles and ensure the stability of the merging process. The lateral merging controller controls the local platoon change the lane to merge into the target platoon safely and smoothly.

IV. SELF-ORGANIZED LOCAL PLATOON FORMATION CONTROL

When multiple connected and automated vehicles merge into the same target platoon, they can merge in sequence to decompose the multi-vehicle merging problem into multiple single-vehicle merging problems. However, this method of

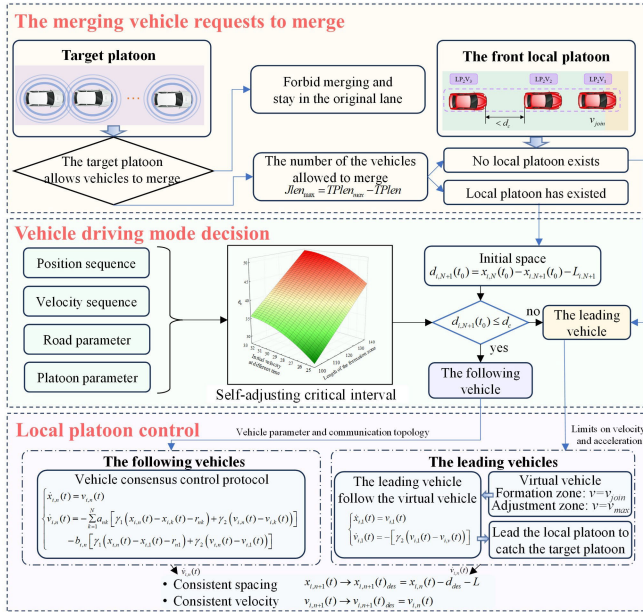


Fig. 3. Schematic diagram of vehicle self-organizing formation control.

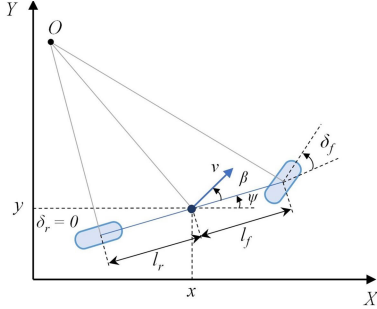


Fig. 4. The bicycle kinematic model.

merging one vehicle at a time is time-inefficient. Even if the target platoon creates multiple gaps at the same time, the target platoon needs to split in multiple places, which results in poor platoon stability and low merging efficiency. Therefore, we propose a vehicle self-organizing formation control strategy based on dynamic self-adjusting critical interval and platoon vehicle consistency control protocol, as shown in Fig. 3. The merging vehicle determines the driving mode as the leading vehicle or the following vehicle based on the dynamic self-adjusting critical interval, and then the platoon vehicle consistency control protocol adjusts the velocity and longitudinal spacing of the local platoon vehicles to be consistent. Eventually, the merging vehicles self-organize into local platoons and complete the merging process in the form of a platoon, which reduces the number of splits of target platoon and improves merging efficiency.

A. Vehicle Model

Vehicle platoons mostly travel on highways, with high longitudinal velocity and small vehicle spacing. Any small longitudinal error during the control process may lead to serious safety problems, which places high demands on the longitudinal controller [30]. The merging vehicles also need

to change lanes to merge into the target platoon, so lateral control of the merging vehicles is also essential.

Therefore, this paper considers not only the longitudinal motion of vehicles to ensure safety and stability during local platoon formation and vehicle merging, but also the lateral control of the local platoon vehicles to achieve lane change and merging.

Consider the bicycle model as shown in Fig. 4:

$$\begin{cases} \dot{x} = v \cdot \cos(\beta + \psi) \\ \dot{y} = v \cdot \sin(\beta + \psi) \\ \dot{\psi} = \frac{v \cdot \tan(\delta_f)}{l_r + l_f} \\ \dot{v} = u \end{cases} \quad (1)$$

where x and y is the longitudinal and lateral position of the vehicle, v is its velocity, ψ is the yaw angle, l_f and l_r represent the distance from the center of gravity of the vehicle to the front and rear axles, u is the acceleration of the vehicle, δ_f is the steering angle, β is the slip angle, which is calculated by the following formula:

$$\beta = \arctan\left(\frac{l_r}{l_f + l_r} \cdot \tan \delta_f\right) \quad (2)$$

The merging process can be divided into longitudinal alignment maneuvers and lateral lane change maneuvers after longitudinal alignment is completed. The longitudinal alignment maneuver and lateral lane change maneuver are dominated by the longitudinal and lateral model of the vehicle, respectively.

Vehicles in the local platoon and target platoon are equipped with V2V devices, and vehicles communicating with each other in the two platoons can exchange their information, including position, velocity, and acceleration.

B. Self-Organized Local Platoon Strategy

The length of the formation zone and adjustment zone in the merging scenario is determined by the coverage of V2V technology, which can be represented as L_{cd} . The total length of the two zones should be less than L_{cd} . It is worth noting that the formation zone and adjustment zone here are dynamic zones that move forward as the target platoon goes ahead. The adjustment zone is always kept within a distance of $L_{az} = k_2 * L_{cd}$ behind the target platoon, correspondingly, the formation zone is kept within a distance of $L_{fz} = k_1 * L_{cd}$ behind the adjustment zone. k_1 and k_2 are the length coefficients and $k_1 + k_2 \leq 1$. The schematic of local platoon formation is shown in Fig. 5. Merging vehicles can communicate with the vehicles in the local platoon ahead and the target platoon. The target platoon travels at a constant velocity v_{des} . Once the merging vehicles enter the formation zone, they begin to form a local platoon. The specific steps are as follows:

1) *The Merging Vehicle Enters the Formation Zone and Sends An Application to the Target Platoon For Merging into the Platoon:* If the number of target platoon vehicles has reached the maximum number of vehicles in the target platoon or the target platoon does not allow vehicles to join,

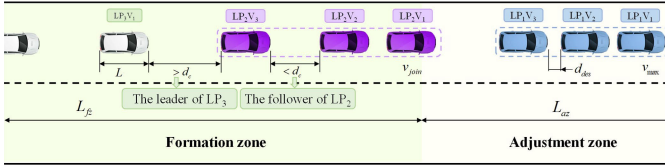


Fig. 5. Local platoon formation process.

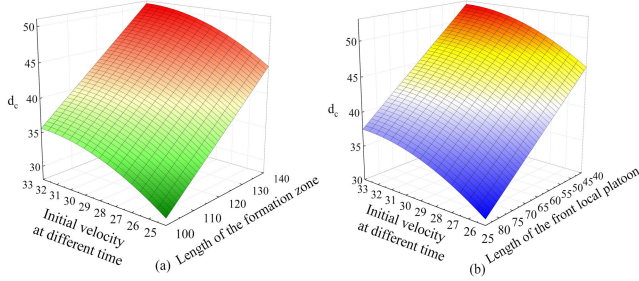


Fig. 6. Dynamic variation of self-adjusting critical interval. (a) The influence of initial velocity at different time and the length of formation zone on self-adjusting critical interval. (b) The influence of initial velocity at different time and the length of front local platoon on self-adjusting critical interval.

the merging vehicle will follow the original route and will not merge.

If the target platoon allows vehicles to join, calculate $Jlen_{max}$, which is the maximum number of vehicles allowed to join the target platoon. $Jlen_{max}$ is calculated by subtracting the maximum number of vehicles from the current number of vehicles in the target platoon, i.e., $Jlen_{max} = TPlen_{max} - TPlen$.

2) *If the Target Platoon Allows Vehicles to Join, Determine Whether There is Already a Local Platoon Ahead:* If there is no local platoon or the local platoon ahead does not allow vehicles to join, i.e., the local platoon ahead reaches $Jlen_{max}$, the merging vehicle becomes the leading vehicle of the next local platoon and travels at a velocity of v_{join} , which is the formation velocity.

If there is a local platoon ahead and it allows vehicle to join, the merging vehicle will be judged whether to join the local platoon ahead based on the initial states of the merging vehicle and the vehicles in the local platoon, which is the next step.

3) *When There is a Local Platoon Ahead, Determine the Driving Mode of the Merging Vehicle:* In the formation zone, vehicles are organized into multiple local platoons according to the initial states. For the current merging vehicle, there can be two different driving modes, i.e., joining the front local platoon to become a following vehicle or becoming the leader of the next local platoon.

The initial spacing between the current merging vehicle and the tail vehicle of the local platoon ahead is defined as:

$$d_{i,N+1}(t_0) = x_{i,N}(t_0) - x_{i,N+1}(t_0) - L \quad (3)$$

where $d_{i,N+1}(t_0)$ is the initial spacing at time t_0 ; $x_{i,N}(t_0)$ is the position of the tail vehicle in the local platoon LP_i at time t_0 ; $x_{i,N+1}(t_0)$ is the position of the merging vehicle; L is the length of the tail vehicle in LP_i . The research focuses on

homogeneous vehicle platoon, so the length of each vehicle is equal.

The dynamic self-adjusting critical interval is solved by the following formula: (4), as shown at the bottom of the next page, where d_c is the dynamic self-adjusting critical interval; N is the number of vehicles in local platoon ahead, $v_{i,N+1}(t_0)$ is the velocity of the merging vehicle; a_{max} is the maximum allowable acceleration given considering vehicle safety and comfort; d_{des} is the desired spacing between vehicles in the platoon. The desired spacing between vehicle i and $i+1$ in the platoon at time t is calculated as follows:

$$d_{des}(t) = \alpha v_{i+1}(t) + \beta \quad (5)$$

where α is the time headway constant and β is the static gap constant.

The part before the equal sign in (4) calculates the distance that the merging vehicle travels more than the local platoon ahead. The part after the equal sign calculates the distance that the merging vehicle needs to travel more than the local platoon when the spacing between the merging vehicle and the tail vehicle of the local platoon ahead reaches the desired spacing. When the two parts are equal, the merging vehicle is just able to close the distance to the vehicle in front of it to d_{des} in the formation zone, i.e., $d_{i,N+1}(t_0) = d_c$. Therefore, when $d_{i,N+1}(t_0) \leq d_c$, the merging vehicle becomes the tail vehicle $LP_i V_{N+1}$ of the local platoon ahead; when $d_{i,N+1}(t_0) > d_c$, the merging vehicle becomes the leading vehicle $LP_{i+1} V_1$ of the next local platoon.

Fig. 6 shows the change of the dynamic self-adjusting critical interval value with the initial velocity of different merging vehicles at different time, the status of local platoon vehicles ahead at different time, and the length of formation zone. As the initial vehicle velocity at different time and the length of the formation zone increase, d_c the also increases, as shown in Fig. 6(a). Similarly, as shown in Fig. 6(b), when the initial vehicle velocity at different time and the length of the local platoon change dynamically, the value of d_c also changes accordingly. Therefore, d_c dynamically adjusts along with the change of parameters to reasonably determine the driving modes of the merging vehicles.

4) *The Merging Vehicle Initially Adjusts its Velocity Within the Formation Zone:* If the merging vehicle becomes the leader of the next local platoon, it adjusts the velocity to drive at a constant velocity of v_{join} . Because the leading vehicle must lead the local platoon to catch up with the target platoon that travels at a velocity of v_{des} as soon as possible, and at the same time take into account the formation of the local platoon, v_{join} satisfies $v_{des} < v_{join} < v_{max}$, v_{max} is the maximum velocity of vehicles allowed on the road.

If the merging vehicle joins the local platoon ahead, it accelerates to catch the tail vehicle ahead. When the spacing between the merging vehicle and the tail vehicle $LP_i V_N$ of the local platoon ahead reaches the desired spacing d_{des} , the velocity of the merging vehicle adjusts to follow $LP_i V_N$. The specific control algorithm is as follows.

C. Local Platoon Control

After the leading vehicle of the local platoon enters the formation zone, it begins to adjust the velocity to lead the local platoon gradually closer to the target platoon. The following vehicles adjust their velocities and spacings following the leading vehicle.

We consider the interaction between vehicles in the platoon and design the platoon vehicle consistency control protocol:

$$\begin{cases} \dot{x}_{i,n}(t) = v_{i,n}(t) \\ \dot{v}_{i,n}(t) = -\sum_{k=1}^N a_{nk} [\gamma_1 (x_{i,n}(t) - x_{i,k}(t) - r_{nk}) + \gamma_2 (v_{i,n}(t) - v_{i,k}(t))] \\ -b_{i,n} [\gamma_1 (x_{i,n}(t) - x_{i,1}(t) - r_{n1}) + \gamma_2 (v_{i,n}(t) - v_{i,1}(t))] \end{cases} \quad (6)$$

where $v_{i,n}(t)$ is the velocity of the vehicle n in local platoon LP_i , a_{nk} is the element of the platoon adjacency matrix, $a_{nk} = 1$ represents that the following vehicle n and vehicle k can communicate with each other and vehicle n can receive the state information sent by vehicle k ; $b_{i,n}$ is the element of the adjacency matrix of the leading vehicle, $b_{i,n} = 1$ represents that the following vehicle n and leading vehicle can communicate with each other and vehicle n can receive the state information sent by leading vehicle; the desired distance between the following vehicle n and vehicle k is represented by r_{nk} ; the desired distance between the following vehicle n and leading vehicle is represented by r_{n1} ; γ_1 and γ_2 are the control gain parameters for platoon and $\gamma_1 > 0$, $\gamma_2 > 0$.

The communication network topology of the platoon is undirected. Two adjacent following vehicles can communicate in two directions to obtain each other's state and each following vehicle can communicate with the leading vehicle to obtain the state of the leading vehicle.

In order to make the velocity of the leading vehicle change smoothly to v_{join} and then to v_{max} in the adjustment zone, suppose there is a virtual vehicle in front of the leading vehicle, and this vehicle always travels at the current desired velocity of the leading vehicle. The leading vehicle always follows the virtual vehicle. Then the control protocol of the leading vehicle can be expressed as:

$$\begin{cases} \dot{x}_{i,1}(t) = v_{i,1}(t) \\ \dot{v}_{i,1}(t) = -[\gamma_2 (v_{i,1}(t) - v_{i,v}(t))] \end{cases} \quad (7)$$

where $v_{i,v}(t)$ are the velocity of the virtual vehicle at time t .

Define the error equation:

$$\begin{aligned} \tilde{x}_{i,n}(t) &= x_{i,n}(t) - x_{i,1}(t) - r_{n1} \\ \tilde{v}_{i,n}(t) &= v_{i,n}(t) - v_{i,1}(t) \end{aligned} \quad (8)$$

where $\tilde{x}_{i,n}(t)$ and $\tilde{v}_{i,n}(t)$ are respectively the position error and velocity error of the vehicle $LP_i V_n$. $x_{i,1}(t)$ and $v_{i,1}(t)$ are respectively the position and velocity of the leading vehicle in LP_i at time t .

Then, equation (6) becomes:

$$\begin{cases} \dot{\tilde{x}}_{i,n}(t) = \tilde{v}_{i,n}(t) \\ \dot{\tilde{v}}_{i,n}(t) = -\sum_{k=1}^N a_{nk} [\gamma_1 (\tilde{x}_{i,n}(t) - \tilde{x}_{i,k}(t)) + \gamma_2 (\tilde{v}_{i,n}(t) - \tilde{v}_{i,k}(t))] \\ -b_{i,n} [\gamma_1 \tilde{x}_{i,n}(t) + \gamma_2 \tilde{v}_{i,n}(t)] \end{cases} \quad (9)$$

When each vehicle in the local platoon satisfies the following conditions, the error approaches zero, and it can be considered that the longitudinal spacing and velocity of the vehicle in the local platoon converge to the consistent:

$$\begin{aligned} x_{i,n+1}(t) &\rightarrow x_{i,n+1}(t)_{des} = x_{i,n}(t) - d_{des} - L \\ v_{i,n+1}(t) &\rightarrow v_{i,n+1}(t)_{des} = v_{i,n}(t) \end{aligned} \quad (10)$$

where $x_{i,n+1}(t)_{des}$ and $v_{i,n+1}(t)_{des}$ are respectively the desired position and desired velocity of the vehicle $LP_i V_{n+1}$ ($\forall n = 1, 2, \dots, N-1$).

Next, we use Lyapunov stability theory to prove that the platoon vehicle consistency control protocol can control the state of the vehicles to be consistent. That is to say, it is proved that when (6) is adopted, if the following vehicles can communicate with each other bidirectionally and the following vehicles can obtain the information of the leading vehicle, then the vehicles can drive stably in a platoon. The distance between the following vehicles and the leading vehicle remains stable, and the velocities of the following vehicles are also consistent with the leading vehicle.

Lemma 1 (LaSalle's Invariance Principle [31]): Let $x(t)$ be a solution of $\dot{x} = f(x)$, $x(0) = x_0 \in R^n$, where $f: U \rightarrow R^n$ is continuous with an open subset U of R^n , and $V: U \rightarrow R^n$ be a locally Lipschitz function such that $D^+ V(x(t)) \leq 0$, where D^+ denotes the upper Dini derivative. Then $\Theta^+(x_0) \cap U$ is contained in the union of all solutions that remain in $S = \{x \in U : D^+ V(x) = 0\}$, where $\Theta^+(x_0)$ denotes the positive limit set.

$$\begin{aligned} &\frac{v_{max}^2 - v_{i,N+1}(t_0)^2}{2a_{max}} + \left(\frac{L f_z - d_c - (N-1) \times L - \sum_{j=1}^{N-2} (x_{i,N-j-1} - x_{i,N-j})}{v_{join} - v_{des}} - \frac{v_{max} - v_{i,N+1}(t_0)}{a_{max}} \right) \\ &\times v_{max} - \frac{L f_z - d_c - (N-1) \times L - \sum_{j=1}^{N-2} (x_{i,N-j-1} - x_{i,N-j})}{v_{join} - v_{des}} \times v_{join} = d_c - L - d_{des} \end{aligned} \quad (4)$$

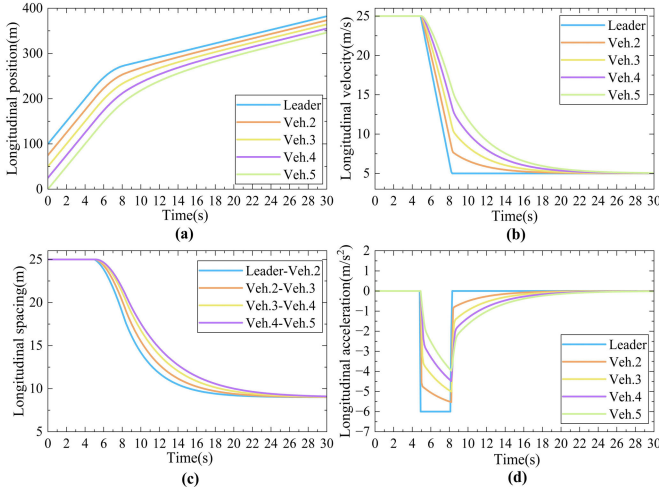


Fig. 7. State curves for vehicles in the case of emergency braking. (a) Longitudinal position. (b) Longitudinal velocity. (c) Longitudinal spacing. (d) Longitudinal acceleration.

Refer to [32] to prove. Firstly, construct the Lyapunov function:

$$\begin{aligned}
 V(t) &= V_1(t) + V_2(t) + V_3(t) \\
 V_1 &= \frac{1}{2} \sum_{i=1}^n \tilde{v}_i(t)^2 \quad V_2 = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \int_0^{\tilde{x}_i(t) - \tilde{x}_j(t)} a_{ij} \gamma_1 s ds \\
 V_3 &= \sum_{i=1}^n \int_0^{\tilde{x}_i(t)} b_i \gamma_1 s ds
 \end{aligned} \quad (11)$$

Considering the derivative of $V_k (k = 1, 2, 3)$ with respect to (9), we can get:

$$\begin{aligned}
 \dot{V}_1 &= \sum_{i=1}^n \tilde{v}_i(t) \dot{\tilde{v}}_i(t) \\
 &= \sum_{i=1}^n \tilde{v}_i(t) \left[- \sum_{j=1}^n a_{ij} (\gamma_1 (\tilde{x}_i(t) - \tilde{x}_j(t)) + \gamma_2 (\tilde{v}_i(t) - \tilde{v}_j(t))) \right. \\
 &\quad \left. - b_i (\gamma_1 \tilde{x}_i(t) + \gamma_2 \tilde{v}_i(t)) \right] \\
 \dot{V}_2 &= \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n a_{ij} (\tilde{v}_i(t) - \tilde{v}_j(t)) \gamma_1 (\tilde{x}_i(t) - \tilde{x}_j(t)) \\
 \dot{V}_3 &= \sum_{i=1}^n b_i \gamma_1 \tilde{v}_i(t) \tilde{x}_i(t)
 \end{aligned} \quad (12)$$

If there is bidirectional communication between the following vehicles, then the topology graph of the following vehicles is undirected, so we can obtain:

$$\begin{aligned}
 \dot{V} &= \dot{V}_1 + \dot{V}_2 + \dot{V}_3 \\
 &= \sum_{i=1}^n \tilde{v}_i(t) \\
 &\quad \times \left[- \sum_{j=1}^n a_{ij} (\gamma_2 (\tilde{v}_i(t) - \tilde{v}_j(t))) - (b_i \gamma_2 \tilde{v}_i(t)) \right]
 \end{aligned}$$

$$\begin{aligned}
 &= - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n a_{ij} \gamma_2 (\tilde{v}_i(t) - \tilde{v}_j(t))^2 - \sum_{i=1}^n b_i \gamma_2 \tilde{v}_i(t)^2 \\
 &\leq - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n a_{ij} (\gamma_2 (\tilde{v}_i(t) - \tilde{v}_j(t)))^2 \leq 0
 \end{aligned} \quad (13)$$

Let $\dot{V} = 0$:

$$- \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n a_{ij} (\gamma_1 (\tilde{v}_i(t) - \tilde{v}_j(t)))^2 - \sum_{i=1}^n b_i \gamma_2 \tilde{v}_i(t)^2 = 0 \quad (14)$$

Equation (14) holds if and only if $\tilde{v}_i = \tilde{v}_j = 0$, $\tilde{v}_i(t) = 0$:

$$\dot{\tilde{v}}_i(t) = - \sum_{j=1}^n a_{ij} \gamma_1 (\tilde{x}_i(t) - \tilde{x}_j(t)) - b_i \gamma_1 \tilde{x}_i(t) = 0 \quad (15)$$

It can be calculated:

$$\begin{aligned}
 \sum_{i=1}^n \tilde{x}_i(t) \dot{\tilde{v}}_i(t) &= - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n a_{ij} \gamma_1 (\tilde{x}_i(t) - \tilde{x}_j(t))^2 \\
 &\quad - \sum_{i=1}^n b_i \gamma_1 \tilde{x}_i(t)^2 = 0
 \end{aligned} \quad (16)$$

From (16), we can get $\tilde{x}_i = \tilde{x}_j = 0, \forall i \neq j$. From Lemma 1, when $t \rightarrow \infty$, $\lim_{t \rightarrow \infty} \tilde{x} = \|x_i - x_{PL} - r_{ip}\| = 0$ and $\lim_{t \rightarrow \infty} \tilde{v} = \|v_i - v_{PL}\| = 0$. The spacing and velocity of the vehicle in the local platoon converge to be consistent, i.e., the state of the vehicles in the local platoon converges to be consistent. The proof is completed.

We also conducted emergency braking experiments to verify the stability of the proposed platoon vehicle consistency control protocol. The velocity of the leading vehicle urgently slowed down from 25m/s to 5m/s. The state curves of the platoon vehicles are shown in Fig. 7. It can be seen that all vehicles can stably follow the leading vehicle without collision.

V. PLATOON COOPERATIVE MERGING CONTROLLERS DESIGN

After the local platoon enters the merging zone, it needs to merge into the target platoon as a whole. When the initial velocities of the local platoon vehicles and the target platoon vehicles are different, the velocities of the platoon vehicles need to be adjusted to be consistent during the process of merging, which may cause wide fluctuations in the velocity of the platoon vehicles. Moreover, the merging of the local platoon requires large platoon gaps, which can also easily cause huge deceleration at the tail of the target platoon [33] and is not conducive to maintaining the stability of the platoon. This paper considers the safe space and the fluctuations in the velocity of the platoons during the merging process, and uses a DMPC algorithm to design two longitudinal cooperative merge controllers, as shown in Fig. 8. By solving two types of constraint optimization problems, smooth and feasible optimal control is obtained, which improves the safety and stability of the merging process. The target platoon splits to create a platoon gap, and the local platoon adjusts its position to align

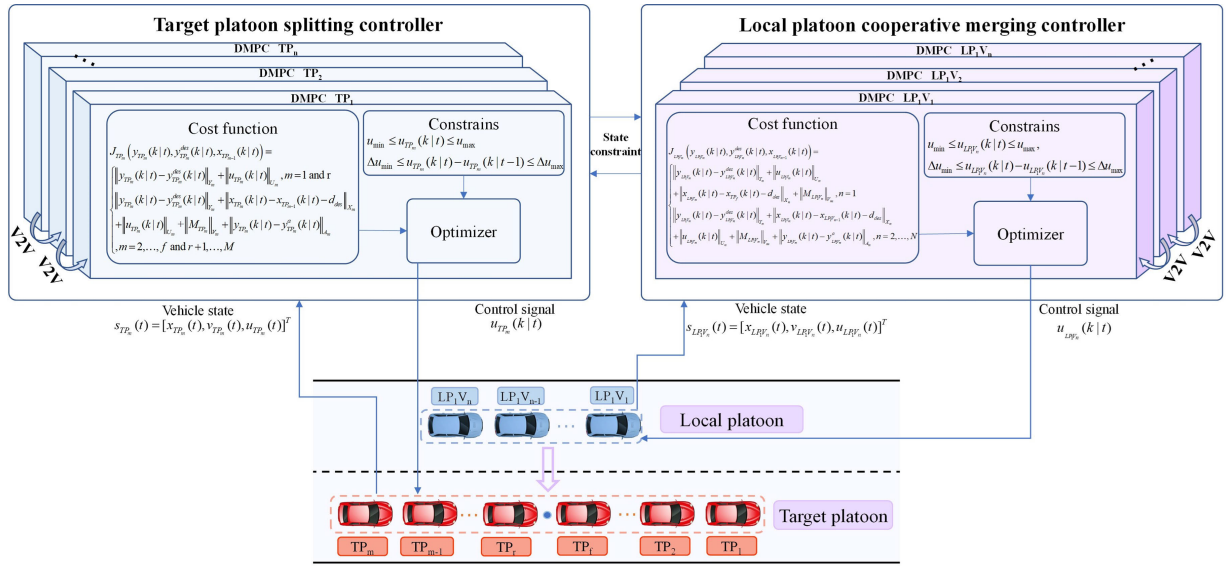


Fig. 8. Schematic diagram of platoon longitudinal cooperative merging control.

with the gap. The local platoon lateral merging controller then controls the local platoon to change lanes to merge into the gap as a whole.

It is worth noting that the local platoon's controller is converted from the vehicle self-organizing formation controller to the cooperative merging controller after the platoon enters the merging zone. The vehicle self-organizing formation controller needs to pass the state information of the vehicles in the local platoon at the current moment to the cooperative merging controller as the initial state, including position, velocity, and acceleration.

A. Platoon Structure Formulation

The process of the local platoon merging into the target platoon is shown in Fig. 9. Two vehicle platoons drive in adjacent lanes. The number of vehicles in the local platoon is N . TP represents the target platoon. The target platoon vehicle in front of the merging point is recorded as TP_f , and the target platoon vehicle behind the merging point is recorded as TP_r .

The communication topology of the two platoons is set as follows. The leader of the local platoon and vehicle TP_f can communicate with each other. The tail vehicle of the local platoon and vehicle TP_r can communicate with each other. Vehicles in the same platoon can communicate with each other. This communication topology makes the communication between the target platoon and the local platoon only need to be completed by the first and last vehicle of the platoon, and the information transmission of the other vehicles is completed by the internal communication of the platoon. This communication topology is simple, and efficiency of the information transmission is high [34].

The discrete-time longitudinal dynamics of platoon vehicles is described by the following double-integral

model:

$$\begin{aligned} x_i(t+1) &= x_i(t) + \tau v_i(t) + \frac{1}{2}\tau^2 u_i(t) \\ v_i(t+1) &= v_i(t) + \tau u_i(t) \end{aligned} \quad (17)$$

where τ ($\tau > 0$) is sampling interval and the initial time is t_0 ; u_i is the control input; $x_i(t)$ and $v_i(t)$ is the position and velocity of the vehicle i at time t .

The state of each vehicle at time t can be expressed as $s_i(t) = [x_i(t), v_i(t), u_i(t)]^T$, and the output of each vehicle at time t is $y_i(t) = [x_i(t), v_i(t)]^T$. Equation (17) can be rewritten as:

$$\begin{aligned} s_i(t+1) &= A s_i(t) + B u_i(t) \\ y_i(t) &= G s_i(t) \end{aligned} \quad (18)$$

where $A = \begin{bmatrix} 1 & \tau & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} \frac{1}{2}\tau^2 \\ \tau \\ 0 \end{bmatrix}$, $G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$; $u_i(t)$ is the control input at time t .

Before the start of the merging process, the position of the vehicle TP_m in the target platoon can be described as:

$$\begin{aligned} x_{TP_m}(t) &= x_{TP_m}(t_0) + \int_{t_0}^t v_{TP_m} dt, \quad (v_{TP_m} = v_{des} : m = 1, \dots, M) \end{aligned} \quad (19)$$

Equation (19) can be discretized as:

$$\begin{aligned} x_{TP_m}(t) &= x_{TP_m}(t_0) + \sum_{i=t_0}^{[(t-t_0)/\tau]} v_{TP_m} \tau, \quad (v_{TP_m} = v_{des} : m = 1, \dots, M) \end{aligned} \quad (20)$$

where $x_{TP_m}(t_0)$ is the initial position of vehicle TP_m at time t_0 ; τ is sampling interval.

Similarly, the position of the vehicle LP_1V_n in the local platoon can be described as:

$$x_{LP_1V_n}(t) = x_{LP_1V_n}(t_0) + \sum_{i=t_0}^{[(t-t_0)/\tau]} v_{LP_1V_n} \tau, (n = 1, \dots, N) \quad (21)$$

After the local platoon merges into the target platoon, the position of the vehicle in the target platoon before the merging point can be described as:

$$\begin{aligned} x_{TP_m}(t) &= x_{LP_1V_1}(t) + d_{TP_f, LP_1V_1}(t) + \sum_{j=m}^{f-m} d_{TP_j, TP_{j+1}}(t) \\ &+ \sum_{j=m}^{f-m+1} L \quad (m = 1, \dots, f) \end{aligned} \quad (22)$$

where $d_{TP_j, TP_{j+1}}$ is the spacing between vehicle TP_j and vehicle TP_{j+1} ; L is the length of the vehicle.

The position of the vehicle in the target platoon after the merging point can be described as:

$$\begin{aligned} x_{TP_m}(t) &= x_{LP_1V_N}(t) - d_{LP_1V_N, TP_r}(t) - \sum_{j=r}^{m-1} d_{TP_j, TP_{j+1}}(t) \\ &- \sum_{j=r}^{m-1} L \quad (m = r, \dots, M) \end{aligned} \quad (23)$$

The new vehicle platoon formed after the local platoon merges into the target platoon is the merged platoon. We hope that vehicles in the merged platoon will maintain the same and stable distance and drive at a constant velocity, so the position of target platoon vehicles in the merged platoon is the desired position of the vehicles during the merging process. Therefore, the desired position of vehicles in the target platoon is:

$$x_{TP_m}^{des} = \begin{cases} x_{LP_1V_1} + d_{des} + L, m = f \\ x_{TP_f}^{des} + \sum_{j=f-m}^f (d_{des} + L), m = 1, \dots, f-1 \\ x_{LP_1V_N} - d_{des} - L, m = r \\ x_{TP_r}^{des} - \sum_{j=m-r}^m (d_{des} + L), m = r+1, \dots, M \end{cases} \quad (24)$$

Similarly, the position of local platoon vehicles in the merged platoon is the desired position of the vehicles during the merging process:

$$x_{LP_1V_n}^{des} = \begin{cases} x_{TP_f} - d_{des} - L, n = 1 \\ x_{LP_1V_1}^{des} - \sum_{i=1}^{n-1} (d_{des} + L), n = 2, \dots, N \end{cases} \quad (25)$$

The desired velocity of the vehicle remains v_{des} during the merging process. When the states of vehicles in two platoons

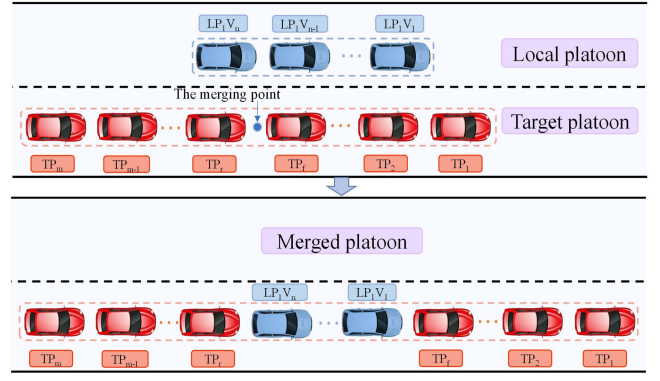


Fig. 9. Scenario of local platoon merging into the target platoon.

satisfy (26) during the merging process, it marks that the local platoon and target platoon merge as the merged platoon, and the merging process ends.

$$\begin{cases} \lim_{t \rightarrow t_{final}} \|v_j(t) - v_{des}\| \leq \delta_v \\ \lim_{t \rightarrow t_{final}} \|x_j(t) - x_j^{des}(t) - d_{des}\| \leq \delta_x \end{cases}, \quad j = 1, \dots, M \text{ and } 1, \dots, N \quad (26)$$

where t_{final} is the time when the merging process ends; δ_v and δ_x are the velocity and position error, ideally the error is zero, i.e., $\delta_v = \delta_x = 0$.

By designing the controllers to control the vehicle to reach the desired position and velocity, the local platoon can be merged into the target platoon. The vehicle control tasks in the merging process are divided into three types: control the splitting of the target platoon to create a platoon gap, control the local platoon to align with the gap and change the lane to merge into the target platoon. Next, we will design the cooperative merging controllers.

B. Target Platoon Splitting Controller

The state of the vehicle in the target platoon at time t is $s_{TP_m}(t) = [x_{TP_m}(t), v_{TP_m}(t), u_{TP_m}(t)]^T$. The desired state of the vehicle in the target platoon at time t is $s_{TP_m}^{des}(t) = [x_{TP_m}^{des}(t), v_{TP_m}^{des}(t), u_{TP_m}^{des}(t)]^T$. The desired output is $y_{TP_m}^{des}(t) = G s_{TP_m}^{des}(t)$. The cost function of each vehicle is defined as: (27), as shown at the bottom of the next page, where $Y_m \in \mathbb{R}^2$, $X_m \in \mathbb{R}$, $U_m \in \mathbb{R}$, $V_m \in \mathbb{R}$ and $A_m \in \mathbb{R}$ are weight matrices, and they are positive symmetric matrices. Y_m is a penalty for output error. X_m is a penalty for spacing error. U_m is a penalty for control input, which can keep the control input as low as possible. V_m is a penalty for velocity fluctuations. A_m is a penalty for deviating from the assumed output. $k = 1, \dots, N_p$ and N_p is the predictive horizon.

M_{TP_m} is the velocity adjustment factor [35], it is calculated by the velocity difference between the current vehicle and the vehicle in front of it. The difference of the velocity of adjacent vehicles is introduced into the cost function, which can accurately describe the velocity fluctuation of the preceding vehicle relative to the following vehicle and reduce the velocity fluctuations.

For the first vehicles TP_1 and TP_r of the two parts after the target platoon splits, their driving modes in the platoon are the leading vehicles, so the cost function only needs to consider the smaller output error and the lower control input. For the remaining vehicles in the target platoon, their driving modes in the platoon is the following vehicles, so the distance and velocity error also need to be considered.

The desired position of the vehicle in the target platoon at each moment can be solved by (24):

$$x_{TP_m}^{des}(k|t) = \begin{cases} x_{LP_1V_1}(k|t) + d_{des} + L, m = f \\ x_{TP_f}^{des}(k|t) + m(d_{des} + L), m = 1, \dots, f-1 \\ x_{LP_1V_N}(k|t) - d_{des} - L, m = r \\ x_{TP_r}^{des}(k|t) - (m-r)(d_{des} + L), \\ m = r+1, \dots, M \end{cases} \quad (28)$$

The constrained optimization problem of the vehicle in the target platoon is:

$$\begin{aligned} & \text{Minimize } \sum_{k=1}^{N_p} J_{TP_m} \\ & u_{\min} \leq u_{TP_m}(k|t) \leq u_{\max}, \\ & \Delta u_{\min} \leq u_{TP_m}(k|t) - u_{TP_m}(k|t-1) \leq \Delta u_{\max}. \end{aligned} \quad (29)$$

where $u_{\min}$ and $u_{\max}$ are the minimum and maximum control input constraints; $\Delta u_{\min}$ and $\Delta u_{\max}$ are the maximum and minimum values of the change between two adjacent control inputs.

C. Local Platoon Longitudinal Merging Controller

Similar to the target platoon, the state of the vehicle in the local platoon at time t is $s_{LP_1V_n}(t) = [x_{LP_1V_2}(t), v_{LP_1V_n}(t), u_{LP_1V_n}(t)]^T$. The desired state of the vehicle in the local platoon at time t is $s_{LP_1V_n}^{des}(t) = [x_{LP_1V_n}^{des}(t), v_{LP_1V_n}^{des}(t), u_{LP_1V_n}^{des}(t)]^T$. The desired output is $y_{LP_1V_n}^{des}(t) = G s_{LP_1V_n}^{des}(t)$. The cost function of each vehicle is defined as: (30), shown at the bottom of the next page, where Y_n , X_n , U_n , V_n and A_n are weight matrices which have the same meaning as in (27). $k = 1, \dots, N_p$ and N_p is the predictive horizon. $M_{LP_1V_n}$ is the velocity adjustment factor same as the definition in (27).

The cost function of the local platoon vehicles needs to take into account the smaller control output error and the lower control input. The leader of the local platoon must lead the platoon to merge into the target platoon, therefore, its cost

Algorithm 1 DMPC for Longitudinal Cooperative Merging Controllers

```

1. Initialize the assumed state of vehicle  $j$  for the first computation:
   
$$\begin{cases} x_j(p|0) = x_j(p-1|0) + v \times \tau \\ v_j(p|0) = v_j(p-1|0) \\ u_j(c|0) = 0 \end{cases},$$

    $j \in \{m|m=1, \dots, M\} \text{ or } \{n|n=1, \dots, N\}$ 
    $p = 2, \dots, N_p, c = 1, \dots, N_c$ 
2. for  $t=t_0:\infty$  do
3.   while  $j < M$  or  $j < N$  do
4.     SolveOptimizationProblem(Minimize  $\sum J_i(y_j, y_j^{des})$ )
5.     SetOptimalControlSequence( $u_j^*(c|t)$ )
6.     ApplyControlQuantity( $u_j^*(1|t)$ )
7.     UpdateAssumedState:
        $s_j^a(k+1|t) = A \cdot s_j^a(k|t) + B \cdot u_j^*(k|t)$ 
        $y_j^a(t) = g \cdot s_j^a(t)$ 
    $k = 1, 2, \dots, N_p - 1$ 
8.   end
9.    $t = t + \tau$ 
10.  If  $isMergingFinish=true$  then
11.    break for;
12.  end if
13. end for

```

function also needs to consider the spacing and the velocity difference of TP_f to control the position and vehicle velocity adjustment of local platoon. For following vehicles in the local platoon, the distance and velocity error also need to be considered.

The desired position of the vehicle in the local platoon at each moment can be solved by (25):

$$x_{LP_1V_n}^{des}(k|t) = \begin{cases} x_{TP_f}(k|t) - d_{des} - L, n = 1 \\ x_{LP_1V_1}^{des}(k|t) - n(d_{des} + L), n = 2, \dots, N \end{cases} \quad (31)$$

The constrained optimization problem of the vehicle in the local platoon is:

$$\begin{aligned} & \text{Minimize } \sum_{k=1}^{N_p} J_{LP_1V_n} \\ & u_{\min} \leq u_{LP_1V_n}(k|t) \leq u_{\max}, \\ & \Delta u_{\min} \leq u_{LP_1V_n}(k|t) - u_{LP_1V_n}(k|t-1) \leq \Delta u_{\max}. \end{aligned} \quad (32)$$

where $u_{\min}$, $u_{\max}$, $\Delta u_{\min}$ and $\Delta u_{\max}$ are the same as in (29).

Equations (29) and (32) consider both the control input and the change of the control input, i.e., the control input constraints. The constraints ensure that the control input remains within a reasonable range and limit the change between two

$$\begin{aligned} & J_{TP_m} \left(y_{TP_m}(k|t), y_{TP_m}^{des}(k|t), x_{TP_{m-1}}(k|t) \right) \\ & = \begin{cases} \left\| y_{TP_m}(k|t) - y_{TP_m}^{des}(k|t) \right\|_{Y_m} + \left\| u_{TP_m}(k|t) \right\|_{U_m}, m = 1 \text{ and } r \\ \left\| y_{TP_m}(k|t) - y_{TP_m}^{des}(k|t) \right\|_{Y_m} + \left\| x_{TP_m}(k|t) - x_{TP_{m-1}}(k|t) - d_{des} \right\|_{X_m} \\ + \left\| u_{TP_m}(k|t) \right\|_{U_m} + \left\| M_{TP_m} \right\|_{V_m} + \left\| y_{TP_m}(k|t) - y_{TP_m}^a(k|t) \right\|_{A_m} \\ m = 2, \dots, f \text{ and } r+1, \dots, M \end{cases} \end{aligned} \quad (27)$$

adjacent control inputs within a certain range, which avoids serious acceleration and deceleration of the vehicles. The velocity adjustment factors in (27) and (30) can maintain low velocity fluctuations of the vehicle. The input constraints and velocity adjustment factors make the vehicle velocity change smoothly, reduce the probability of collision, and improve the safety and stability of the merging process.

The algorithm steps of longitudinal cooperative merging controllers are shown in Algorithm 1.

D. Local Platoon Lateral Merging Controller

When the longitudinal states of the local platoon vehicles are adjusted well, they need to change the lane to merge into the target platoon. Firstly, it is necessary to plan a lateral trajectory for each vehicle in the local platoon, and then a controller is designed to control the steering wheel angle of the vehicle to achieve the tracking of the lateral trajectory.

Bézier curves are parametric curves that are widely used in the field of intelligent driving due to their low computational cost and geometric and curvature continuity. They are smooth curves generated within the convex envelope formed by the control points, which meet the safety and comfort of the vehicle during lane changing. Therefore, the lane change trajectories are planned for the local platoon vehicles based on quintic Bézier curves, as shown in Fig. 10. w is the width of the lane. p_0 to p_5 are the six control points of the curve. p_0 is the lane change starting point and p_5 is the lane change ending point. In order to ensure that the curvature at the starting point and ending point of the lane change is zero, the first three and last three control points are on the same straight line with equal longitudinal interval which is D , and $D > w$ [36].

The basic expression of the quintic Bézier curve is:

$$B(t) = \sum_{i=0}^5 p_i \cdot \binom{5}{i} \cdot (1-t)^{5-i} \cdot t^i, t \in [0, 1] \quad (33)$$

where p_i is the control point used to generate the curve, t is the parameter for curve construction.

Take the starting point of the lane change as the origin to establish a coordinate system, the x-axis indicates the longitudinal position of the vehicle, and the y-axis indicates the transverse position of the vehicle. Then the longitudinal coordinates of the six control points that we have selected are 0, D , $2D$, $3D$, $4D$ and $5D$ respectively. Substituting into (33) gives:

$$B(t)_x = 5Dt, t \in [0, 1] \quad (34)$$

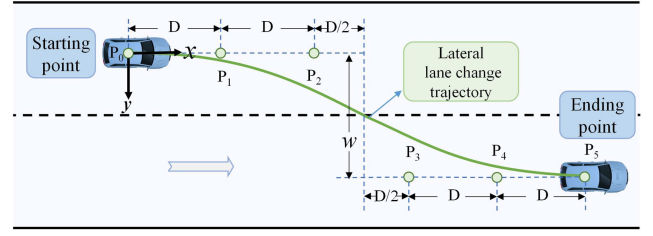


Fig. 10. Trajectory planning on lane change.

Substituting $t = B_x/5D$ into (33) yields:

$$B(B_x)_y = \sum_{i=0}^5 p_i \cdot \binom{5}{i} \cdot (1 - B_x/5D)^{5-i} \cdot (B_x/5D)^i B_x \in [0, 5D] \quad (35)$$

where B_x is the longitudinal position of the vehicle, $B(B_x)_y$ is the desired lateral position of the vehicle when the longitudinal position is B_x .

At this point, the expression of the planned lane change trajectory is obtained in the established lane change coordinate system. In other words, the desired lateral path is a function of the longitudinal position of the vehicle.

In order to track the planned lateral lane change trajectory, a lateral PID controller is used to control the vehicle's steering wheel angle:

$$\delta(t) = K_p e(t)_y + K_i \int e(t)_y dt + K_d \frac{de(t)_y}{dt} \quad (36)$$

$$e(t)_y = y_{des} - y(t)$$

where $\delta(t)$ is the steering angle at time t , and $e(t)_y$ is the lateral error which shows the deviation of the vehicle's lateral position from the desired one. $e(t)_y = y_{des} - y(t)$ and y_{des} is defined by (35). K_p , K_i and K_d are the lateral PID controller gains.

By planning a lateral lane change trajectory for each vehicle in the local platoon and controlling its steering wheel angle, the local platoon can change the lane to merge into the target platoon.

VI. SIMULATION RESULTS

In this section, we conduct simulation experiments on the proposed multi-vehicle cooperative control strategy for platoon formation. In a typical scenario where multi-vehicle merge into the target platoon cooperatively, the effectiveness of the

$$J_{LP_1 V_n} \left(y_{LP_1 V_n}(k|t), y_{LP_1 V_n}^{des}(k|t), x_{LP_1 V_{n-1}}(k|t) \right) = \begin{cases} \left\| y_{LP_1 V_n}(k|t) - y_{LP_1 V_n}^{des}(k|t) \right\|_{Y_n} + \left\| u_{LP_1 V_n}(k|t) \right\|_{U_n} \\ + \left\| x_{LP_1 V_n}(k|t) - x_{TP_f}(k|t) - d_{des} \right\|_{X_n} + \left\| M_{LP_1 V_n} \right\|_{V_n}, n = 1 \\ \left\| y_{LP_1 V_n}(k|t) - y_{LP_1 V_n}^{des}(k|t) \right\|_{Y_n} + \left\| x_{LP_1 V_n}(k|t) - x_{LP_1 V_{n-1}}(k|t) - d_{des} \right\|_{X_n} \\ + \left\| u_{LP_1 V_n}(k|t) \right\|_{U_n} + \left\| M_{LP_1 V_n} \right\|_{V_n} + \left\| y_{LP_1 V_n}(k|t) - y_{LP_1 V_n}^a(k|t) \right\|_{A_n}, n = 2, \dots, N \end{cases} \quad (30)$$

TABLE I
SIMULATION PARAMETERS

Parameters	Value	Description
L_{fz}	140m	The length of the formation zone
L_{az}	60m	The length of the adjustment zone
v_{des}	25m/s	The desired velocity of the platoon
v_{join}	30m/s	The formation velocity of the platoon
v_{max}	33m/s	The maximum velocity
u_{max}, u_{min}	3m/s ² , -6m/s ²	The minimum and maximum acceleration
$\Delta u_{max}, \Delta u_{min}$	0.5m/s ² , -0.5m/s ²	The minimum and maximum acceleration difference between two adjacent control inputs
L	5m	The length of the vehicle
d_{safe}	20m	The desired safe distance
τ	0.1s	simulation horizon
γ_1, γ_2	0.5, 1	Control parameters of consensus protocols
N_p	20	Predictive horizon
$AorY, X, U, V$	$\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}, 5, 10, 5$	DMPC control parameters
w	4	The width of the lane
K_p, K_i, K_d	0.5, 0.4, 0.15	The lateral PID controller gains.

vehicle self-organizing formation control and the cooperative merging controllers is verified. Then, comparative experiments are conducted with the position-prior strategy [15] and LMPC (Laguerre-based Model predictive control) strategy [37].

A. Simulation Parameters and Scenarios

The simulation scenario is shown in Fig. 11. The main advantage of platoon is reduced fuel consumption. The advantages of platoon can be better utilized when driving at high speed. Therefore, vehicle platoons usually travel on highway and we conducted experiments in high-speed scenarios. V2V dedicated short-range communication (DSRC) [38] is used between vehicles in the target platoon and the merging vehicles. At present, the coverage range of this technology is about 300 m. In order to ensure the stability of communication, the total length of the formation zone and adjustment zone is set to 200 m. In order to make the merging vehicles form a local platoon reasonably, the length of the formation zone should be longer than that of the adjustment zone, which is 140 m and 60 m respectively. Moreover, on the highway, the velocity of vehicles is usually between 90 km/h and 120 km/h, and the distance required to adjust the velocity of vehicles to be consistent is relatively small, so the length distribution of the above formation zone and adjustment zone is reasonable. It is worth noting that the formation zone and adjustment zone here are dynamic zones. They move forward as the target platoon goes ahead and remain in the zone behind the target platoon.

The maximum allowable driving velocity of vehicles is set to 33 m/s because the maximum velocity on the highway is not allowed to exceed 120 km/h. Correspondingly, the desired velocity and formation velocity are set to 25 m/s and 30 m/s respectively. The desired safe spacing is calculated by (5) and $\alpha = 0.8$, $\beta = 5$. In consideration of vehicle performance and

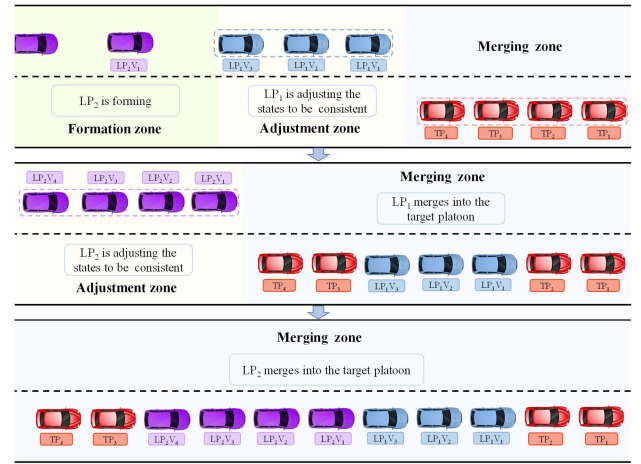


Fig. 11. Schematic diagram of simulation process.

the fluctuations of velocity, we also limit the acceleration. The detailed simulation parameters are shown in Table I.

In the simulation, a total of eleven vehicles are set up, of which four vehicles are in the target platoon and the remaining eleven vehicles are merging vehicles. Merging vehicles appear in the formation zone behind the target platoon with random velocity ranging from 18 m/s to 33 m/s. Since we mainly focus on the process of vehicles merging into the platoon, we start the simulation when the vehicles enter the formation zone and do not pay attention to the driving conditions before entering.

B. Results and Analysis

The longitudinal position of vehicles from entering the formation zone to merging into the target platoon is shown in Fig. 12. Seven merging vehicles enter the formation zone with random velocity at 0 s, 5 s, 11 s, 18 s, 25 s, 30 s, and 35.7 s after the simulation starts. According to the velocity and position of each vehicle, the first vehicle and the fourth vehicle are calculated by (4) to become the leading vehicles of two local platoons respectively. The two local platoons completely enter the merging zone at 44 s and 64 s, and the target platoon splitting controller and the local platoon longitudinal merging controller coordinate the three platoons to adjust their longitudinal position to the desired position. The longitudinal positions of the three platoons are adjusted well at 59 s and 87 s, and then the local platoon lateral merging controller control the local platoons to change lanes to merge into the target platoon. The lateral position of the two local platoon vehicles when they change lanes to merge into the target platoon is shown in Fig. 13. It can be seen that the local platoon lateral merging controller closely follows the desired lateral trajectory designed by the planner and safely executes the lane change maneuver. Finally, the two local platoons merge into the target platoon safely and stably.

The state curves of the merging vehicles in the formation zone and adjustment zone are shown in Fig. 14. Fig. 14(a) shows the distance of the merging vehicles relative to the merging zone, i.e., the distance relative to the target platoon.

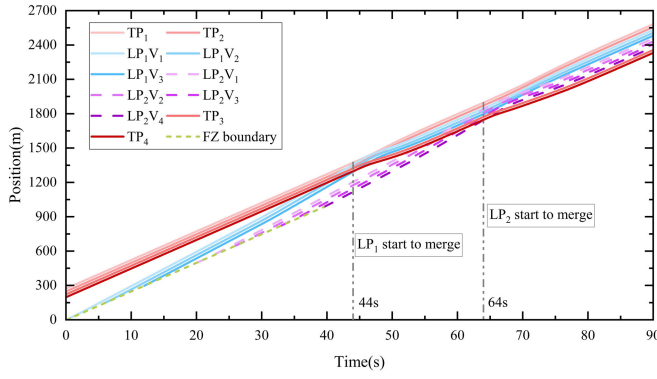


Fig. 12. Longitudinal position trajectory for eleven vehicles.

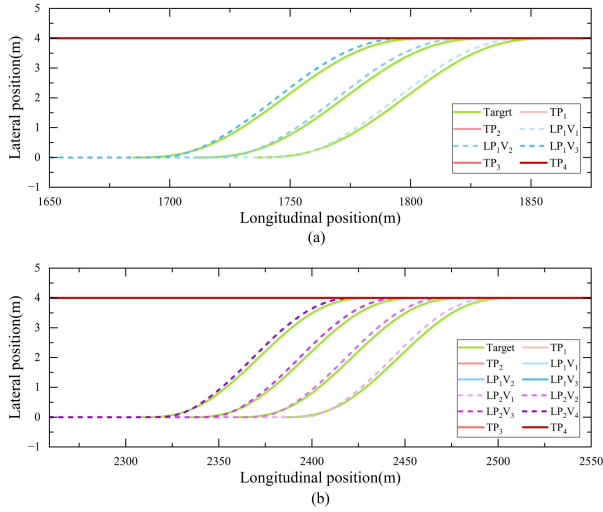


Fig. 13. Lateral position trajectory for local platoon vehicles. (a) Lateral position trajectory during LP1 merging process. (b) Lateral position trajectory during LP2 merging process.

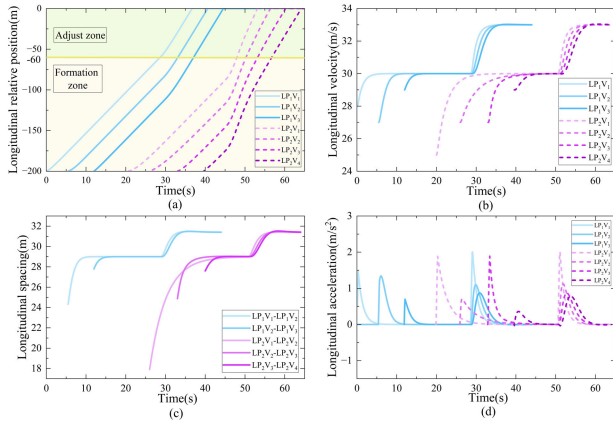


Fig. 14. State curves for merging vehicles in formation and adjustment zones. (a) Longitudinal relative position. (b) Longitudinal velocity. (c) Longitudinal spacing. (d) Longitudinal acceleration.

When the relative position is -200 m, vehicles enter the formation zone, and when the relative position is zero, vehicles exit the adjustment zone. Before LP_1 enters the merging zone, the target platoon travel at the desired velocity, and the vehicles in LP_1 approach the merging zone smoothly and stably. Before

LP_2 enters the merging zone, LP_1 has already started to merge into the target platoon and the velocity of the target platoon changes, so the relative position of the vehicles in LP_2 also changes. The longitudinal velocity curves of the merging vehicles are shown in Fig. 14(b). The initial velocity of the vehicles when they enter the formation zone is between 25 m/s and 33 m/s, which adjusts to 30 m/s in the formation zone and accelerates to 33 m/s after entering the adjustment zone. Fig. 14(c) is the longitudinal spacing curves of vehicles in two local platoons. The spacing of vehicles in the platoon adjusts to 29 m in the formation zone. In the adjustment zone, the spacing adjusts to 31.4 m due to the increase in vehicle velocity. LP_1 and LP_2 enter the adjustment zone and start to accelerate at around 29s and 51s respectively, so the acceleration and spacing of the vehicles fluctuate slightly here. In the end, the velocity and spacing of vehicles in both platoons remain consistent. The merging vehicles form local platoons and enter the merging zone.

Because most of the control methods of vehicle merging into the platoon are aligning the longitudinal positions of vehicles first before changing lanes to merge, and most of the research focuses on the longitudinal control in the merging process. In other words, the longitudinal control of the merging process is relatively more difficult. Therefore, the comparative experiments are aimed at longitudinal control.

The longitudinal state curves of LP_1 and LP_2 using three different control strategies to merge into the target platoon are shown in Fig. 15 and Fig. 16 respectively. When the multi-vehicle cooperative merging control strategy is adopted, for the longitudinal position in Fig. 15(a), the velocity of vehicles TP_0 and TP_1 first increases and then decreases to the desired velocity, and the vehicles move forward some distance relative to the local platoon. The velocity of vehicles TP_2 and TP_3 first decreases and then increases to the desired velocity, and the vehicles move some distance backward relative to the local platoon. TP_0 , TP_1 , TP_2 and TP_3 jointly create platoon gap for the local platoon to merge. Similarly, in Fig. 16(a), TP_4 and the vehicles before it move forward to create a platoon gap for the local platoon together with TP_5 and the vehicles after it. At the same time, LP_1 and LP_2 merge into the target platoon as a whole. During the merging process, the velocity and acceleration (Fig. 15(b)(d), Fig. 16(b)(d)) of the vehicles maintain within a limited range, which indicates that multi-vehicle cooperative merging control strategy not only ensures the safety, and also improves the stability of the platoons. The longitudinal spacing between two adjacent vehicles in the platoon is shown in Fig. 15(c) and Fig. 16(c). Finally, the spacing between vehicles converges to 25 m, and the velocity of all vehicles also converges to 25 m/s and remains stable.

Under the same initial position and velocity, the merging vehicles can also merge into the target platoon by using the position-prior strategy and LMPC strategy, and the longitudinal velocity and spacing of the vehicles reach the same and remain stable. However, it takes about 15 s and 23 s to control LP_1 and LP_2 to align with the target platoon gap using the multi-vehicle cooperative merging control strategy, while the position-prior strategy and LMPC strategy takes about 20 s, 18 s and 30 s, 27s, which indicates that the proposed strategy

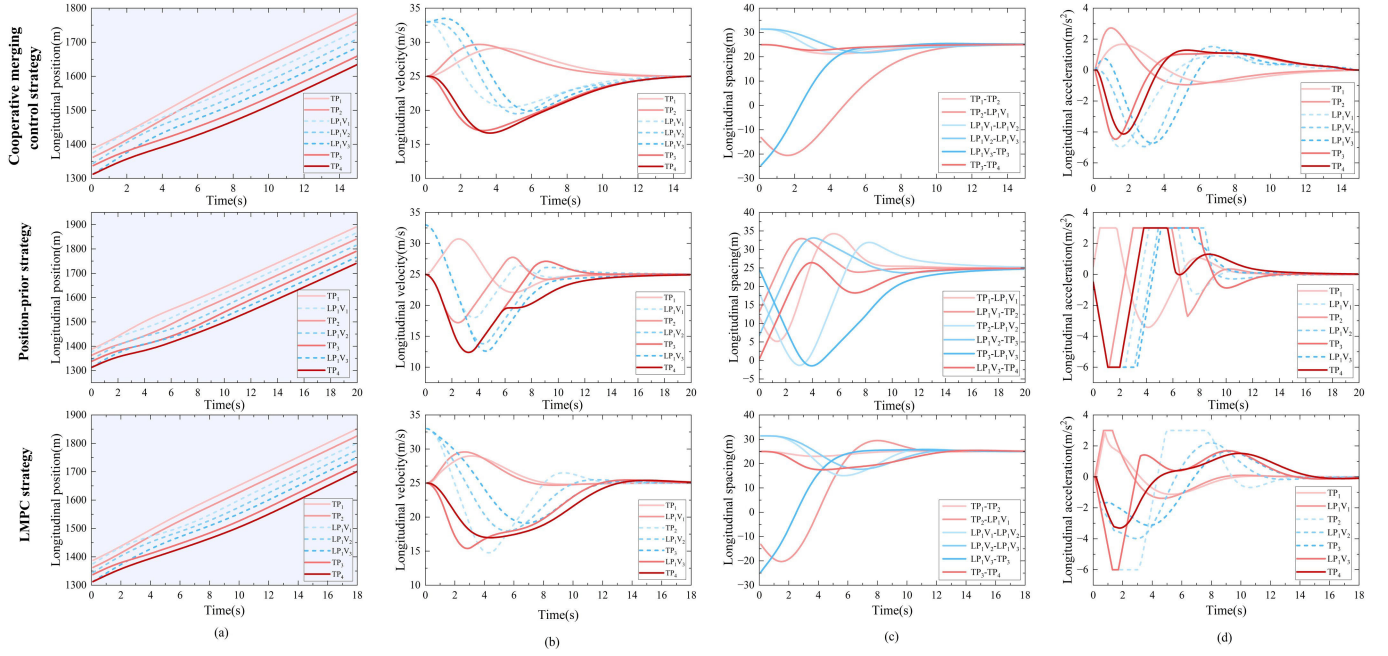


Fig. 15. Simulation results during LPI merging process with three different methods. (a) Longitudinal position. (b) Longitudinal velocity. (c) Longitudinal spacing. (d) Longitudinal acceleration.

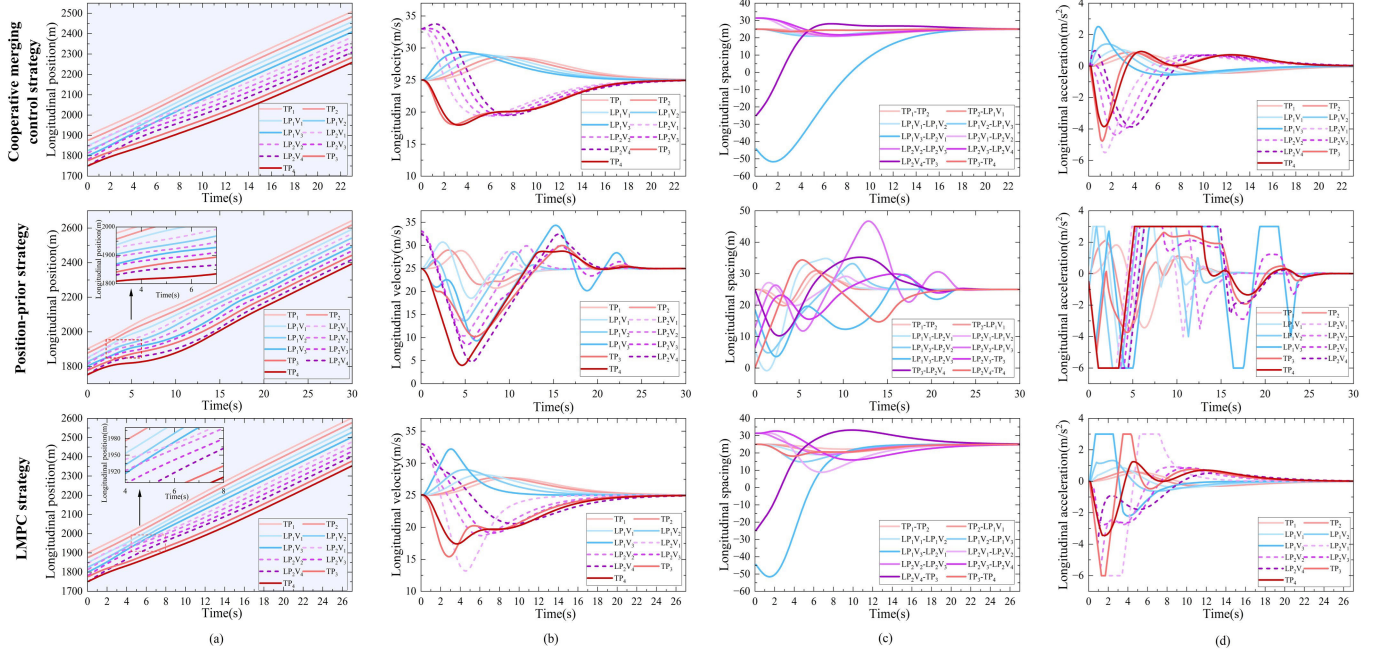


Fig. 16. Simulation results during LP2 merging process with three different methods. (a) Longitudinal position. (b) Longitudinal velocity. (c) Longitudinal spacing. (d) Longitudinal acceleration.

significantly improves the merging efficiency. Moreover, it can be seen from Fig. 15(b)(c)(d) and Fig. 16(b)(c)(d) that there are many times of wide fluctuations in velocity, acceleration and longitudinal spacing of vehicles during the process of merging using the position-prior strategy, and the overshooting is very large. In Fig. 16(c), the maximum longitudinal spacing between two adjacent vehicles even reaches 47m. The acceleration fluctuation is also very large during the process of merging using LMPC strategy, and the accelerations of

many vehicles reach the acceleration limit. And the changes of velocity and longitudinal spacing are also not as smooth as the proposed strategy. In Fig. 16(a), both of the two comparison methods will result in the situation that the front vehicle decelerates urgently and the rear vehicle fails to avoid, which will shorten the distance between the two vehicles sharply, and the risk of collision is very high.

The longitudinal velocity and spacing errors calculated by (26) of the three strategies during the merging process

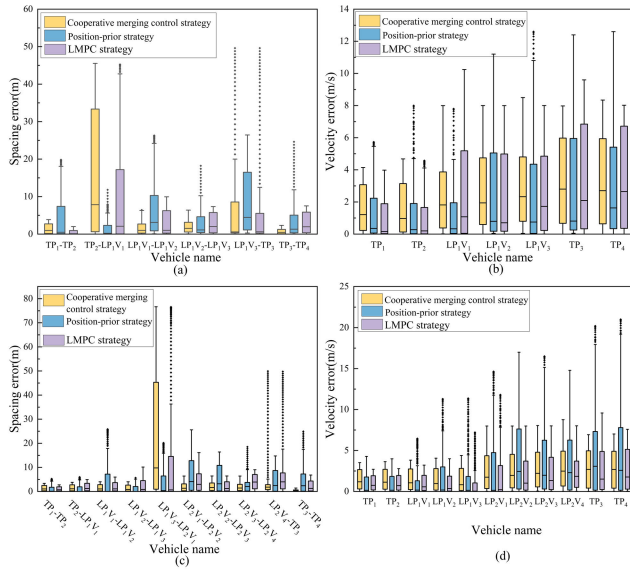


Fig. 17. Spacing and velocity errors using three different strategies. (a) Spacing error for seven vehicles. (b) Velocity error for seven vehicles. (c) Spacing error for eleven vehicles. (d) Velocity error for eleven vehicles.

TABLE II
THE VARIANCES OF VEHICLE ACCELERATION DURING
LP1 MERGING PROCESS

Vehicle	TP ₁	TP ₂	LP ₁ V ₁	LP ₁ V ₂	LP ₁ V ₃	TP ₃	TP ₄
Proposed strategy	0.52	0.86	3.02	3.45	3.30	2.24	2.29
Position-prior strategy	2.01	4.89	3.46	6.65	5.21	6.70	4.28
LMPC strategy	0.71	1.06	8.22	3.81	2.47	3.69	1.97

are shown in Fig. 17. The variances of the acceleration are shown in Table II and Table III. Combined with the spacing and velocity curves in Fig. 15 and Fig. 16, the velocity and spacing errors are smaller when using the cooperative merging control strategy, and converge faster and more smoothly. Most of the velocity and spacing errors of the vehicles when using the cooperative merging control strategy are smaller than when using the position-prior strategy and LMPC strategy. In Fig. 17(a)(c), there are some spacing errors that are larger when the cooperative merging control strategy and LMPC strategy are adopted which is because the position-prior strategy chooses the merging position closer to the merging vehicle so that the initial error is small. In Fig. 17(b)(d), the velocity distributions of some vehicles are more dispersed with the cooperative merging control strategy than with the position-prior strategy and LMPC strategy, which is because their velocities converge slowly and oscillate around the desired velocity for a long time under the position-prior strategy. Although the spacing and velocity of vehicles can approach the desired value quickly when LMPC is adopted, the convergence speed is slow in the late stage and the spacing and velocity of vehicles are near the desired value for a long time.

TABLE III
THE VARIANCES OF VEHICLE ACCELERATION DURING
LP2 MERGING PROCESS

Vehicle	TP ₁	TP ₂	LP ₁ V ₁	LP ₁ V ₂	LP ₁ V ₃	LP ₂ V ₁	LP ₂ V ₂	LP ₂ V ₃	LP ₂ V ₄	TP ₃	TP ₄
Proposed strategy	0.14	0.16	0.19	0.26	0.42	2.26	1.89	1.73	1.84	1.19	1.07
Position-prior strategy	1.01	0.96	2.99	5.10	4.06	6.10	8.95	6.60	3.18	8.04	6.24
LMPC strategy	0.08	0.08	0.13	0.26	1.32	5.57	1.91	1.43	0.88	2.54	1.11

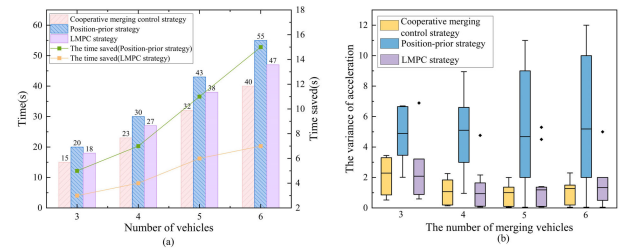


Fig. 18. Merging time and the variances of acceleration for different number of vehicles using three different strategies. (a) Merging time. (b) The variances of acceleration.

From Table II and Table III, the variances of acceleration when using the proposed strategy are generally smaller than that of the position-prior strategy and LMPC strategy, so the merging process is more stable. In general, the smooth convergence process of the cooperative merging control strategy reduces the risk of collision and the merging time, and the merging process is safer and more stable.

To assess the impact of the proposed strategy on traffic flow, we calculated the longitudinal average velocity during the merging process using three different strategies, i.e. dividing the longitudinal travel distance by the longitudinal travel time. The average velocity of the seven vehicles during LP1 merging is 24.14 m/s when using the cooperative merging control strategy, while it is and is the average speed of the seven vehicles is 23.37 m/s and 23.83 m/s when using the position-prior strategy and LMPC strategy, respectively. The average velocity of the eleven vehicles during LP2 merging is 24.65 m/s when using the cooperative merging control strategy, while it is and is the average speed of the eleven vehicles is 23.31 m/s and 24.38 m/s when using the position-prior strategy and LMPC strategy, respectively. It can be seen that the average velocity is higher when using the cooperative merging control strategy, which is more conducive to improving traffic efficiency and increasing traffic flow.

We also conducted comparative experiments of three to six vehicles merging into the same target platoon using three strategies. The merging time and the variances of acceleration are shown in Fig. 18. The time required by the cooperative merging control strategy is less than that of the position-prior strategy and LMPC strategy, which improves the merging efficiency. And as the number of vehicles increases, the advantage becomes more obvious. For the cooperative merging

control strategy, the variances of acceleration always keep in a low range, but when the position-prior strategy is adopted, they will fluctuate greatly with the change of vehicle number, and the stability of the platoon is poor. When LMPC strategy is adopted, the acceleration variance of individual vehicles is too large, which is not conducive to the stability of the platoon.

In summary, the proposed multi-vehicle cooperative merging control strategy can reduce the complexity of multi-vehicle merging into the same platoon and enable vehicles to merge into the target platoon efficiently, safely and smoothly.

VII. CONCLUSION

Aiming at the problems of low merging efficiency, poor platoon stability, and high collision risk when multiple connected and automated vehicles merge into the same target platoon, we propose a multi-vehicle self-organized cooperative control strategy for platoon formation. Based on the dynamic self-adjusting critical interval and the platoon vehicle consistency control protocol, the vehicle self-organizing formation control strategy is designed, which divides the merging vehicles into several local platoons according to their initial states, reduces the complexity of multi-vehicle merging into the same platoon and improves the merging efficiency. Considering the safe space and the changes in the velocity of the platoons, two cooperative merging controllers based on DMPC are designed to solve two types of constrained optimization problems. The target platoon is split to create a platoon gap, and the local platoon merges into the target platoon safely and smoothly.

The simulation results show that the cooperative merging control strategy can reduce the complexity of multi-vehicle merging into the same platoon and enable vehicles to merge into the target queue efficiently, safely, and stably. In the future, we will focus on the cooperative merging of heterogeneous vehicle platoons.

REFERENCES

- [1] P. Zhang et al., "Joint optimization of platoon control and resource scheduling in cooperative vehicle-infrastructure system," *IEEE Trans. Intell. Vehicles*, vol. 8, no. 6, pp. 3629–3646, Jun. 2023.
- [2] M. Hu, C. Li, Y. Bian, H. Zhang, Z. Qin, and B. Xu, "Fuel economy-oriented vehicle platoon control using economic model predictive control," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 20836–20849, Nov. 2022.
- [3] M. Sun, A. Al-Hashimi, M. Li, and R. Gerdes, "Impacts of constrained sensing and communication based attacks on vehicular platoons," *IEEE Trans. Veh. Technol.*, vol. 69, no. 5, pp. 4773–4787, May 2020.
- [4] Q. Li, Z. Chen, and X. Li, "A review of connected and automated vehicle platoon merging and splitting operations," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 12, pp. 22790–22806, Dec. 2022.
- [5] J. Wu, Y. Wang, and C. Yin, "Curvilinear multilane merging and platooning with bounded control in curved road coordinates," *IEEE Trans. Veh. Technol.*, vol. 71, no. 2, pp. 1237–1252, Feb. 2022.
- [6] J. Ploeg, B. T. M. Scheepers, E. van Nunen, N. van de Wouw, and H. Nijmeijer, "Design and experimental evaluation of cooperative adaptive cruise control," in *Proc. 14th Int. IEEE Conf. Intell. Transp. Syst. (ITSC)*, Oct. 2011, pp. 260–265.
- [7] J. Ploeg, N. van de Wouw, and H. Nijmeijer, "Lp string stability of cascaded systems: Application to vehicle platooning," *IEEE Trans. Control Syst. Technol.*, vol. 22, no. 2, pp. 786–793, Mar. 2014.
- [8] X. Guo, J. Wang, F. Liao, and R. S. H. Teo, "Distributed adaptive integrated-sliding-mode controller synthesis for string stability of vehicle platoons," *IEEE Trans. Intell. Transp. Syst.*, vol. 17, no. 9, pp. 2419–2429, Sep. 2016.
- [9] J. Lan and D. Zhao, "Min-max model predictive vehicle platooning with communication delay," *IEEE Trans. Veh. Technol.*, vol. 69, no. 11, pp. 12570–12584, Nov. 2020.
- [10] G. An and A. Talebpour, "Vehicle platooning control for merge coordination: A hybrid ACC-DMPC approach," in *Proc. IEEE 25th Int. Conf. Intell. Transp. Syst. (ITSC)*, Oct. 2022, pp. 1611–1616.
- [11] M. Cheng, Z. Li, Z. Wang, X. Luo, B. Liu, and H. Hu, "Collaborative control switching strategy for merging into an automated platoon," in *Proc. IEEE 25th Int. Conf. Intell. Transp. Syst. (ITSC)*, Oct. 2022, pp. 3449–3454.
- [12] W. J. Scholte, P. W. A. Zegelaar, and H. Nijmeijer, "A control strategy for merging a single vehicle into a platoon at highway on-ramps," *Transp. Res. C, Emerg. Technol.*, vol. 136, Mar. 2022, Art. no. 103511.
- [13] S. Maiti, S. Winter, L. Kulik, and S. Sarkar, "The impact of flexible platoon formation operations," *IEEE Trans. Intell. Veh.*, vol. 5, no. 2, pp. 229–239, Jun. 2020.
- [14] E. S. Kazerooni and J. Ploeg, "Interaction protocols for cooperative merging and lane reduction scenarios," in *Proc. IEEE 18th Int. Conf. Intell. Transp. Syst.*, Sep. 2015, pp. 1964–1970.
- [15] C. Hidalgo, R. Lattarulo, C. Flores, and J. P. Rastelli, "Platoon merging approach based on hybrid trajectory planning and CACC strategies," *Sensors*, vol. 21, no. 8, p. 2626, Apr. 2021.
- [16] J. Ding, L. Li, H. Peng, and Y. Zhang, "A rule-based cooperative merging strategy for connected and automated vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 8, pp. 3436–3446, Aug. 2020.
- [17] J. Liu, C. Wang, W. Zhao, and Z. Zhang, "Energy-efficient hierarchical on-ramp merging control for autonomous electric vehicles," *IEEE Trans. Transport. Electric.*, 2024.
- [18] R. Wu, H. Jia, Q. Huang, J. Tian, H. Gao, and G. Wang, "Multi-lane unsignalized intersection cooperation strategy considering platoons formation in a mixed connected automated vehicles and connected human-driven vehicles environment," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 2, pp. 1569–1585, Feb. 2024.
- [19] D. Liu, S. Mair, K. Yang, S. Baldi, P. Frasca, and M. Althoff, "Resilience in platoons of cooperative heterogeneous vehicles: Self-organization strategies and provably-correct design," *IEEE Trans. Intell. Vehicles*, vol. 9, no. 1, pp. 2262–2275, Jan. 2024.
- [20] C. Mu, L. Du, and X. Zhao, "Event triggered rolling horizon based systematical trajectory planning for merging platoons at mainline-ramp intersection," *Transp. Res. C, Emerg. Technol.*, vol. 125, Apr. 2021, Art. no. 103006.
- [21] Y. Xue, X. Zhang, Z. Cui, B. Yu, and K. Gao, "A platoon-based cooperative optimal control for connected autonomous vehicles at highway on-ramps under heavy traffic," *Transp. Res. C, Emerg. Technol.*, vol. 150, May 2023, Art. no. 104083.
- [22] A. I. M. Medina, N. van de Wouw, and H. Nijmeijer, "Cooperative intersection control based on virtual platooning," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 6, pp. 1727–1740, Jun. 2018.
- [23] X. Xiong, J. Sha, and L. Jin, "Optimizing coordinated vehicle platooning: An analytical approach based on stochastic dynamic programming," *Transp. Res. B, Methodol.*, vol. 150, pp. 482–502, Aug. 2021.
- [24] Z. Deng, K. Yang, W. Shen, and Y. Shi, "Cooperative platoon formation of connected and autonomous vehicles: Toward efficient merging coordination at unsignalized intersections," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 5, pp. 5625–5639, May 2023.
- [25] S. Karbalaieali, O. A. Osman, and S. Ishak, "A dynamic adaptive algorithm for merging into platoons in connected automated environments," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 10, pp. 4111–4122, Oct. 2020.
- [26] M. Charitidou and D. V. Dimarogonas, "Splitting and merging control of multiple platoons with signal temporal logic," in *Proc. IEEE Conf. Control Technol. Appl. (CCTA)*, Aug. 2022, pp. 1031–1036.
- [27] H. Min, Y. Yang, Y. Fang, P. Sun, and X. Zhao, "Constrained optimization and distributed model predictive control-based merging strategies for adjacent connected autonomous vehicle platoons," *IEEE Access*, vol. 7, pp. 163085–163096, 2019.
- [28] S. Zhou, S. Li, and G. Tan, "Age of information in V2V-enabled platooning systems," *IEEE Internet Things J.*, vol. 11, no. 3, pp. 4015–4028, Feb. 2024.
- [29] J. Hu and G. Feng, "Distributed tracking control of leader-follower multi-agent systems under noisy measurement," *Automatica*, vol. 46, no. 8, pp. 1382–1387, Aug. 2010.
- [30] X. Ge, Q.-L. Han, J. Wang, and X.-M. Zhang, "Scalable and resilient platooning control of cooperative automated vehicles," *IEEE Trans. Veh. Technol.*, vol. 71, no. 4, pp. 3595–3608, Apr. 2022.

- [31] N. Rouche, P. Habets, and M. Laloy, *Stability Theory by Lyapunov's Direct Method*. New York, NY, USA: Springer-Verlag, 1977.
- [32] Z.-H. Guan, F.-L. Sun, Y.-W. Wang, and T. Li, "Finite-time consensus for leader-following second-order multi-agent networks," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 59, no. 11, pp. 2646–2654, Nov. 2012.
- [33] W. J. Scholte, P. W. A. Zegelaar, and H. Nijmeijer, "Gap opening controller design to accommodate merges in cooperative autonomous platoons," *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 15294–15299, 2020.
- [34] O. Pauca, A. Maxim, and C. F. Caruntu, "Communication topologies evaluation for a vehicle merging into a platoon on highway," in *Proc. 30th Medit. Conf. Control Autom. (MED)*, Jun. 2022, pp. 957–962.
- [35] X. Zhuang, J. Zhang, J. Tian, F. Cui, and T. Wang, "Variable time headway spacing strategy for connected vehicles platoon based on sliding mode control," *Phys. A, Stat. Mech. Appl.*, vol. 637, Mar. 2024, Art. no. 129588.
- [36] R. Lattarulo, L. González, E. Martí, J. Matute, M. Marcano, and J. Pérez, "Urban motion planning framework based on N-Bézier curves considering comfort and safety," *J. Adv. Transp.*, vol. 2018, no. 1, Jul. 2018, Art. no. 6060924.
- [37] A. Kazemi, I. Sharifi, and H. A. Talebi, "Longitudinal and lateral control of vehicle platoons using Laguerre-based and robust MPC with merge and exit maneuvers," *Control Eng. Pract.*, vol. 142, Jan. 2024, Art. no. 105737.
- [38] E. Moradi-Pari, D. Tian, M. Bahramgiri, S. Rajab, and S. Bai, "DSRC versus LTE-V2X: Empirical performance analysis of direct vehicular communication technologies," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 5, pp. 4889–4903, May 2023.



Wanzhong Zhao received the B.S. and M.S. degrees in vehicle engineering from Jiangsu University, Zhenjiang, China, in 2004 and 2005, respectively, and the Ph.D. degree in vehicle engineering from Beijing Institute of Technology, Beijing, China, in 2009. He is currently a Professor and the Director with the Department of Vehicle Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing, China. His research interests include intelligent vehicle and vehicle system dynamics.



Jinjiang Liu received the B.S. degree in aircraft power engineering from Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2019, where he is currently pursuing the Ph.D. degree in mechanical engineering. His research interests include intelligent control and intelligent transportation systems.



Mengqi Zhang received the B.S. degree in vehicle engineering from Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2023, where she is currently pursuing the Ph.D. degree with the Department of Mechanical Engineering. Her research interests include intelligent transportation systems and man-machine cooperative driving.



Chunyan Wang received the B.S. and Ph.D. degrees in mechanical engineering from Jilin University, Changchun, China, in 2000 and 2008, respectively. She is currently a Professor with the Department of Vehicle Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing, China. Her research interests include intelligent vehicle and vehicle system dynamics.



Ziyu Zhang received the B.S. and Ph.D. degrees in vehicle engineering from Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2019 and 2024, respectively. He is currently a Post-Doctoral Researcher with the Department of Vehicle Engineering, Nanjing University of Aeronautics and Astronautics. His research interests include man-machine cooperative driving and vehicle system dynamics.